Solving Simple Logarithmic Equations - Nathaniel Juan P

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Student

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Topic: Solving Simple Logarithmic Equations

Status: completed

Lesson Plan

Learning Objectives

- The student will be able to convert logarithmic equations into their equivalent exponential forms.
- The student will be able to solve logarithmic equations containing a single logarithmic term for the unknown variable.
- The student will be able to justify the validity of solutions by checking for extraneous roots within the logarithmic domain.
- The student will be able to apply logarithmic properties to isolate variables in structured algebraic problems.

Lesson Information

Lesson Sections

1. Spiral Review Warm-up (10 minutes)

Teacher Script:

Hi Nathaniel! Great to see you. We're diving into solving equations today, but first, let's recalibrate your 'logarithmic brain' with a quick vault challenge. I've posted four quick tasks on the whiteboard. Two focus on the Change of Base you mastered last time, and two on your log properties. Ready? Go ahead and solve these in the chat or on the board.

Activities:

- *REVIEW: Change of Base - Evaluate log_4(32) using base 2.*
- *REVIEW: Change of Base - Express log_3(10) in terms of common logs (base 10).*
- *REVIEW: Properties - Simplify log(x^2) + log(y) into a single logarithm.*
- *REVIEW: Properties - Expand log(a/b^3).*

2. Introduction & Real-World Connection (5 minutes)

Teacher Script:

Think of a logarithm as a secure digital lock. In cybersecurity, we often use 'logarithmic complexity' to describe how hard it is to crack a password. If we have the 'encrypted' log value, solving the equation is like finding the original key. Today, we learn the 'Inverse Key Method'—turning logs back into exponents to solve for x.

Activities:

- *Visualizing the relationship: $\log_b(x) = y$ vs $b^y = x$*

3. Review of Prior Knowledge (5 minutes)

Teacher Script:

Before we solve, let's refresh the Golden Rule: the argument of a log (the stuff inside) must always be greater than zero. If I solve an equation and get $x = -5$, but the original equation was $\log(x)$, is that a valid solution? No! We call that an extraneous solution. Keep your eyes peeled for those today.

Activities:

- *Defining the domain of $f(x) = \log(x)$.*

4. Problem Introduction (10 minutes)

Teacher Script:

Let's look at our first 'Level 1' lock: $\log_2(x + 3) = 4$. To solve this, we rewrite it. The base is 2, the exponent is 4, and the result is $(x + 3)$. So, $2^4 = x + 3$. What is 2 to the power of 4? Correct, 16. Now we just have $16 = x + 3$. Subtract 3, and $x = 13$. Does $13 + 3 > 0$? Yes! The lock is open.

Activities:

- *Step-by-step demonstration of converting log to exponential form.*
- *Check for extraneous solutions.*

5. Guided Practice (15 minutes)

Teacher Script:

Now it's your turn to drive. I've placed three equations on the board. For the first one, $\log_5(2x - 1) = 2$, tell me: what is my first step? Good. Now, what about $\log_x(64) = 3$? Here the base is the variable! Finally, let's try a common log: $\log(x^2 + 9) = 1$. Remember, if there's no base written, it's base 10.

Activities:

- *Problem 1: $\log_5(2x - 1) = 2$*
- *Problem 2: $\log_x(64) = 3$*
- *Problem 3: $\log(x^2 + 9) = 1$*

6. Independent Practice (15 minutes)

Teacher Script:

You're a natural at this. I'm setting a timer for 12 minutes. There are four 'Hard' level challenges on the screen. Some might require you to use the log properties we reviewed in the warm-up before you convert. If you see two logs on one side, condense them first! Go!

Activities:

- *Solve: $\log_2(x) + \log_2(x - 2) = 3$*
- *Solve: $\log_3(x^2 - 8x) = 2$*
- *Solve: $2 \log_5(x) = \log_5(9)$*
- *Solve: $\log(x) + \log(x+3) = 1$*

7. Consolidation & Reflection (5 minutes)

Teacher Script:

Fantastic work today, Nathaniel. You've mastered the art of converting log equations to solve them. Remember: 1. Condense logs if needed. 2. Rewrite as an exponent. 3. Solve for x. 4. Always check if x makes the log argument negative! Before you go, please log in to https://learn.algonova.id/login to access your post-session notes and practice.

Activities:

- *Recap of the 4-step checklist.*
- *Self-rating of confidence (1-5).*

Homework

Medium Questions:

1. Solve: $\log_3(x - 5) = 2$
2. Solve: $\log_2(3x + 2) = 5$
3. Solve: $\log_x(100) = 2$
4. Solve: $\log(2x) = 3$

Hard Questions:

1. Solve: $\log_2(x) + \log_2(x - 6) = 4$
2. Solve: $\log_3(x^2 - 7) = 2$
3. Solve: $\log(x + 21) + \log(x) = 2$
4. Solve: $\log_5(x + 1) - \log_5(x - 1) = 1$
5. Solve: $2 \log_3(x) - \log_3(x - 2) = 2$
6. Solve for x: $\log_2(\log_3(x)) = 1$

Answer Key:

`log_3(x-5)=2 => 3^2=x-5 => 9+5=x => x=14`
`log_2(3x+2)=5 => 2^5=3x+2 => 32=3x+2 => 30=3x => x=10`
`log_x(100)=2 => x^2=100 => x=10 (base must be positive)`
`log(2x)=3 => 10^3=2x => 1000=2x => x=500`
`log_2(x(x-6))=4 => x^2-6x=16 => x^2-6x-16=0 => (x-8)(x+2)=0. x=8 (x=-2 is extraneous)`
`log_3(x^2-7)=2 => x^2-7=9 => x^2=16 => x=4, x=-4 (both valid as (-4)^2-7>0)`
`log(x^2+21x)=2 => x^2+21x-100=0 => (x+25)(x-4)=0. x=4 (x=-25 is extraneous)`
`log_5((x+1)/(x-1))=1 => (x+1)/(x-1)=5 => x+1=5x-5 => 6=4x => x=1.5`
`log_3(x^2/(x-2))=2 => x^2/(x-2)=9 => x^2=9x-18 => x^2-9x+18=0 => (x-6)(x-3)=0. x=6, x=3`
`log_2(log_3(x))=1 => 2^1=log_3(x) => 3^2=x => x=9`
REVIEW: Change of Base - `log_9(27) = log_3(27)/log_3(9) = 3/2 = 1.5` (Medium)
REVIEW: Change of Base - `log_4(x) = 2.5 => 4^2.5 = (2^2)^2.5 = 2^5 = 32` (Medium)
REVIEW: Change of Base - Solve for x: `(log 8) / (log 2) = x => log_2(8) = 3` (Hard)
REVIEW: Change of Base - Evaluate `(log_5 12) * (log_12 25)` using base change: `(log 12 / log 5) * (log 25 / log 12) = log 25 / log 5 = 2` (Hard)
REVIEW: Properties - `log_a(x) + 2 log_a(y) - log_a(z) = log_a(xy^2/z)` (Medium)
REVIEW: Properties - If `log 2 = a` and `log 3 = b`, find `log 18`: `log(2*3^2) = a + 2b` (Medium)
REVIEW: Properties - Solve `log_2(x^2) = (log_2 x)^2`: `2 log_2 x = (log_2 x)^2`. Let `u = log_2 x`. `2u = u^2 => u(u-2)=0`. `log_2 x = 0 (x=1)` or `log_2 x = 2 (x=4)` (Hard)
REVIEW: Properties - Simplify `10^(2 log x + log 3)`: `10^(log(3x^2)) = 3x^2` (Hard)

Parent Reminder

Nathaniel's Math Progress: Logarithmic Equations

Today Nathaniel worked on 'cracking the code' of logarithmic equations! He learned how to convert complex logs into exponential forms to solve for unknowns. We also focused on 'extraneous solutions'—tricky answers that look right but don't fit the rules of math. He's doing great with these abstract concepts!